

Abstract

Cold plasma (CP) is an eco-friendly method to enhance seed germination, yet outcomes remain difficult to predict due to complex seed-plasma-environment interactions. This study introduces the first machine learning framework to forecast germination uplift in soybean, barley, sunflower, radish, and tomato under dielectric barrier discharge (DBD) plasma. Among the models tested (GB, XGB, ET, and hybrids), Extra Trees (ET) performed best ($R^2 = 0.919$; RMSE = 3.21; MAE = 2.62), further improving to $R^2 = 0.925$ after feature reduction. Engineering analysis revealed a hormetic response: negligible effects at 7 kV or 200 s, maximum germination at 7–15 kV for 200–500 s, and reduced germination beyond 20 kV or prolonged exposures. Discharge power was also a dominant factor, with germination rate maximizing at 100W with low exposure time. Species and cultivar-level predictions showed radish (MAE = 1.46) and soybean (MAE = 2.05) were modeled with high consistency, while sunflower remained slightly higher variable (MAE = 3.80).

Introduction

Agricultural production is highly depended on successful conversion on seed into strong plants. Conventionally, seed are primed with chemicals and hydropriming, which has limitations such as chemical residuals, high water demand, and inconsistent performance under different environmental conditions¹. In recent years, Cold Plasma (CP) has emerged as an eco-friendly, non-thermal seed priming technology which enhances seed coat properties and activates various physiological responses to improve germination without compromising viability^{1,2}.

However, as germination responses vary widely depending on seed traits, plasma parameters, and environmental conditions, it is challenging to find an optimal priming condition through conventional experimentation alone². Machine learning offers a powerful framework to capture these complex interactions among biological and engineering drives by enabling prediction and optimization on non-linear surface^{1,2}.

This study develops a machine learning-based predictive framework to forecast CP-induced germination enhancement across multiple crops, aiming to support data-driven optimization of plasma priming for sustainable seed and crop improvement technologies.

Materials and Methods

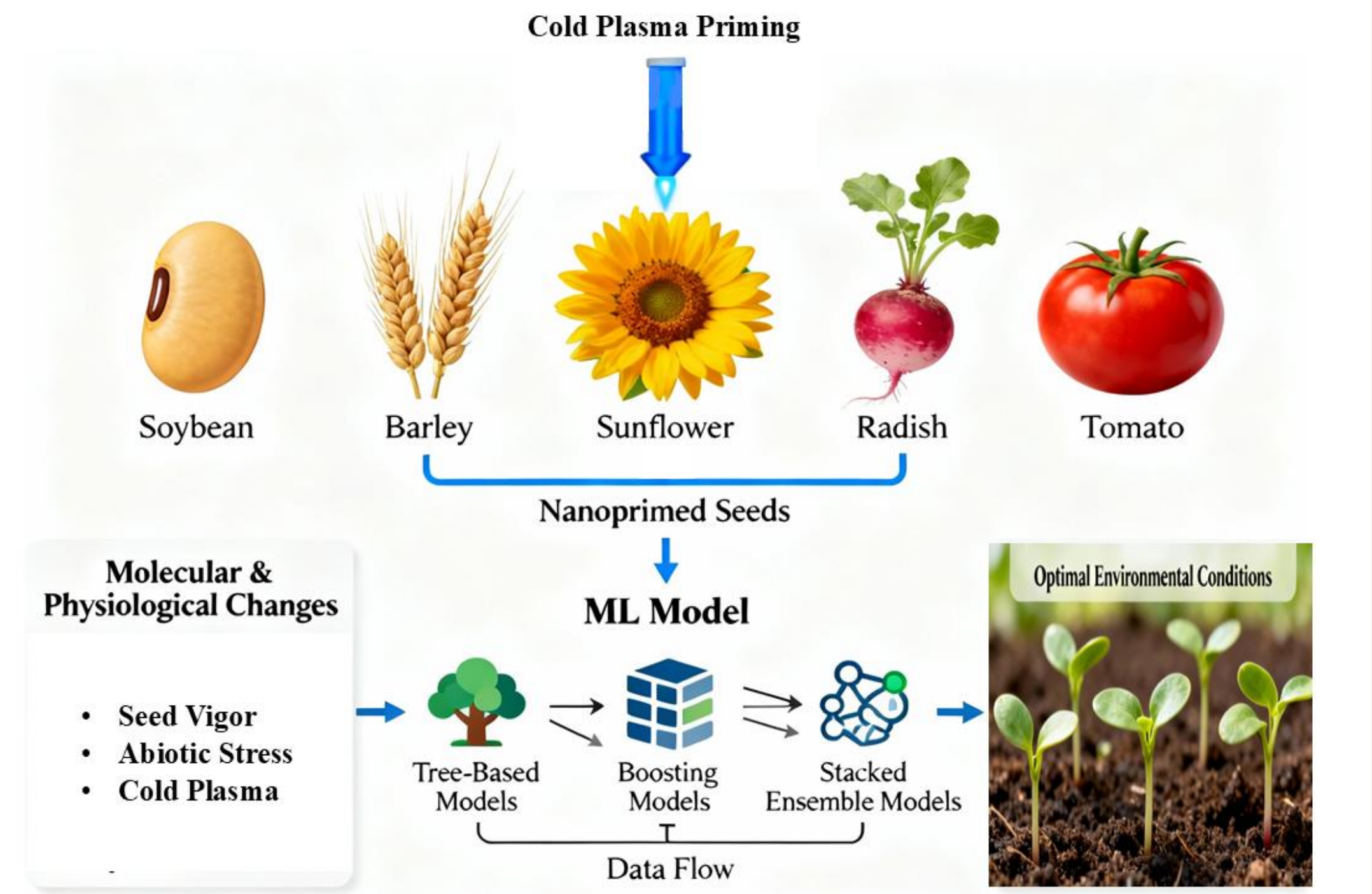


Figure 1: Materials and Methodology

Results and Discussion

Overall model performance: ET achieved the best overall performance with an R^2 of 93%.

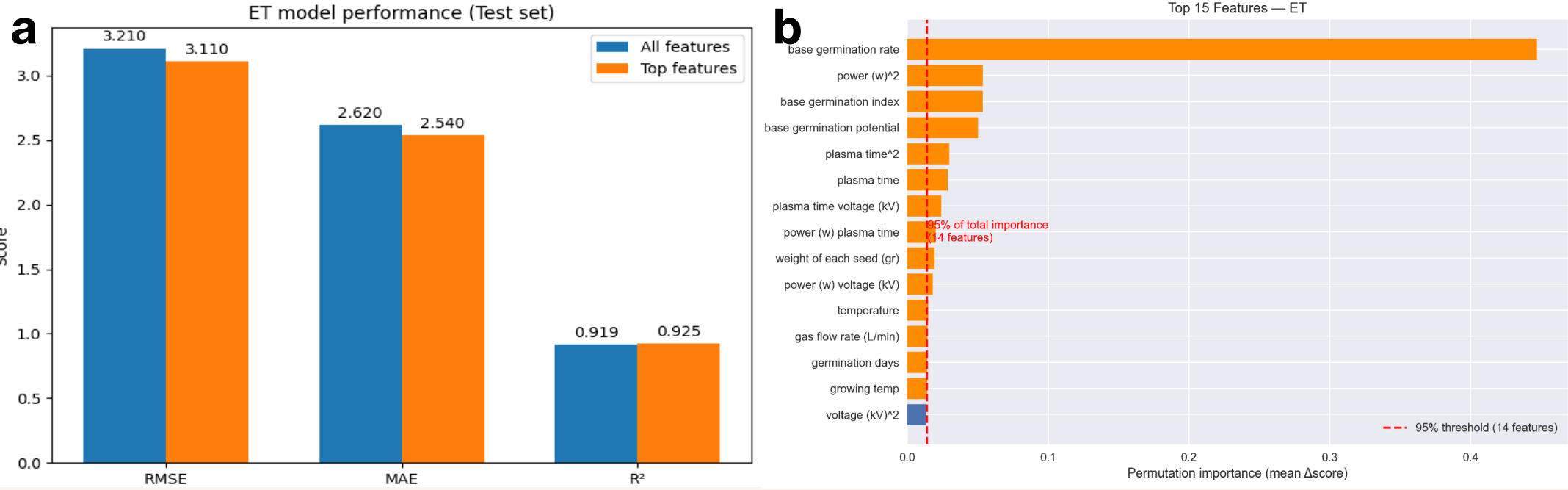


Figure 2: a) Feature Importance, b) All feature and feature reduced model comparison

Biological response pattern: High-vigor seeds showed negligible or negative response. Seeds with <80% baseline germination showed up to 20–30% uplift.

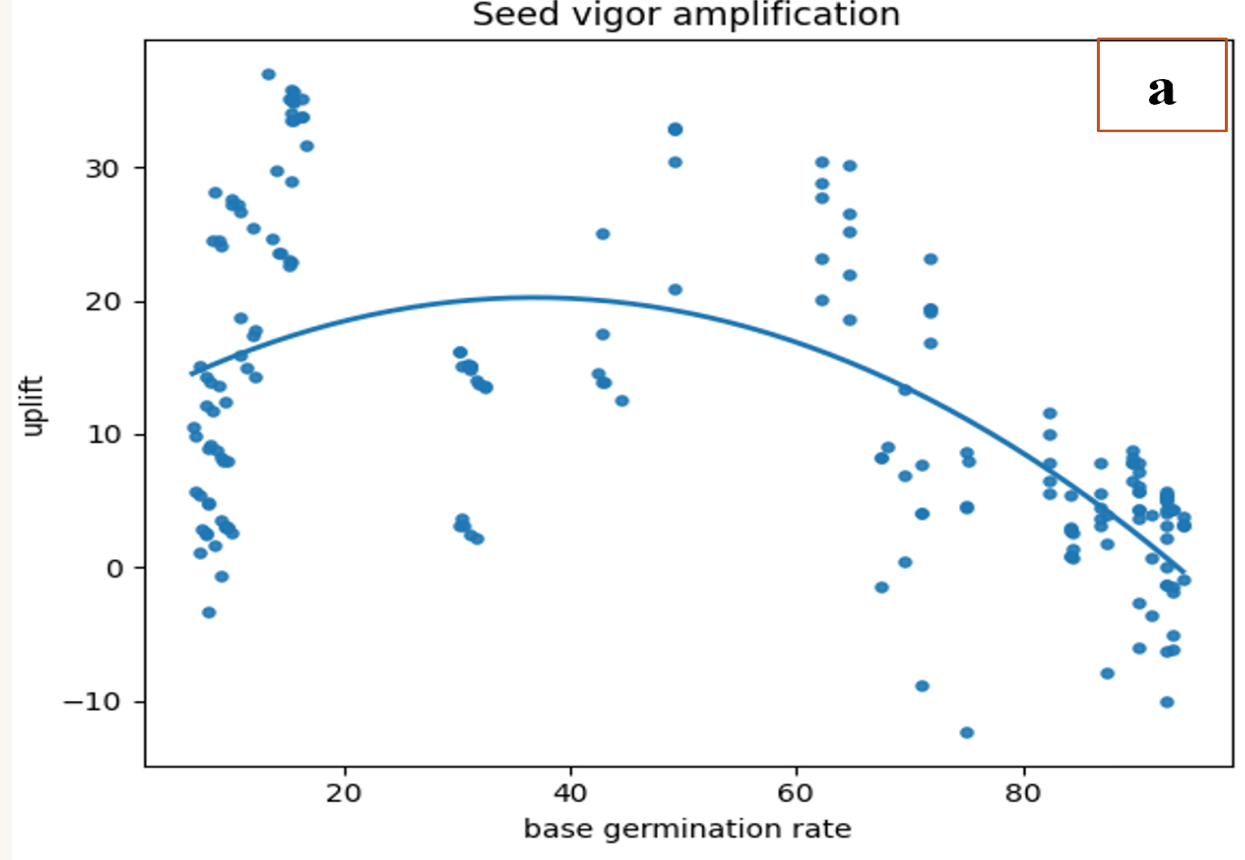


Figure 3: a) Cold plasma reaction based on base seed vigor.

Plasma operating window:

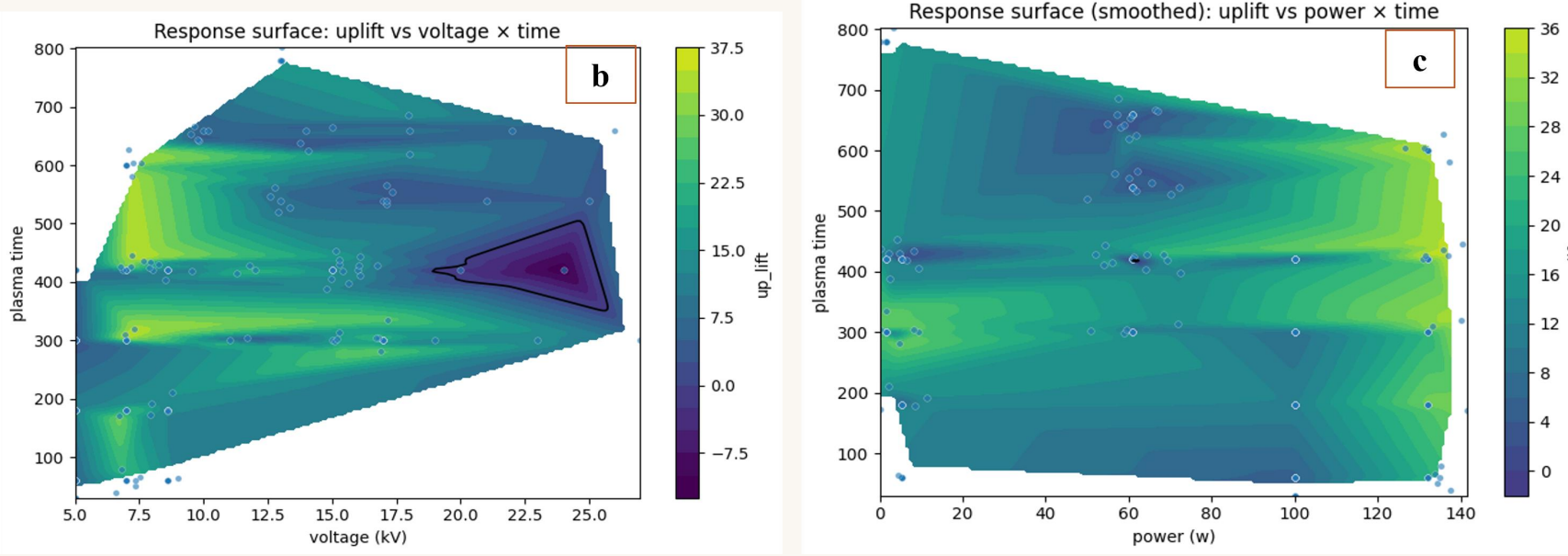


Figure 4: Effect of b) plasma voltage, and c) plasma power on germination rate.

SHAP analysis:

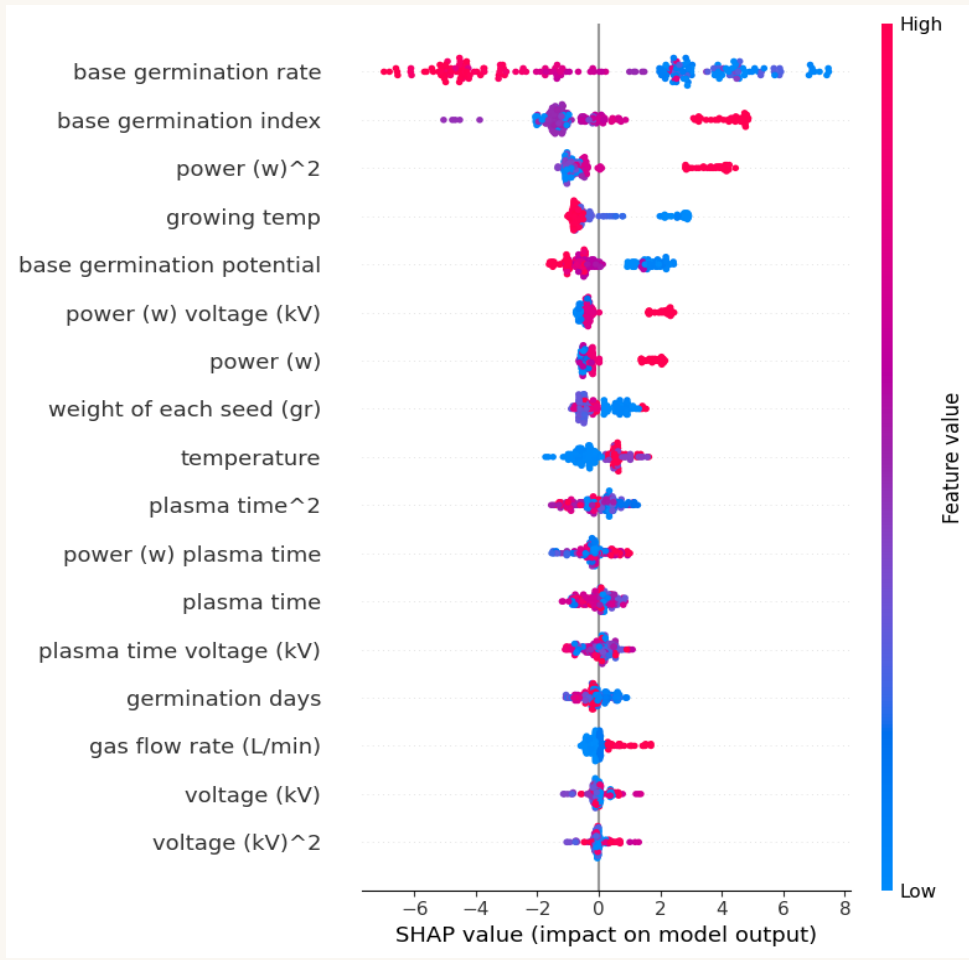


Figure 5: SHAP Analysis

References

- Shahabi, Z. M., Nasibi, F., & Noori, H. (2025). Cold plasma technology as a pre-treatment for seed priming enhances germination and reduces salinity stress in *Prosopis koelziana*. Scientific Reports, 15, 26250. <https://doi.org/10.1038/s41598-025-11637-z>
- Anu, Manonmani, V., Jerlin, R., Vanitha, S., Sivakumar, R., Preetha, G., & Jha, R. (2025). Cold plasma: A green technology for improving legume productivity. Recent Advances in Agriculture by Young Minds – II, 12(sp1). <https://doi.org/10.14719/pst.7993>